

P2P Single Channel LoRa Gateway



Applications

- Building Automation & Monitoring
- Smart Metering & Lighting Solutions
- Smart Waste Management
- Industrial Monitoring
- Inventory Management
- Smart Flood Sensors
- Smart Street Lighting
- Smart City Network
- Environmental Monitoring
- Smart Agriculture

Features

- *Range upto 5 km (Line of Sight) and over 1 km in dense urban environments
- Support upto 30 node on single gateway (*based on field scenario this will change)
- Long-range wireless communication using LoRa RF technology
- Point-to-Point (P2P) communication architecture
- Single frequency channel operation
- Low power consumption design
- Industrial-grade hardware for harsh environments
- High receiver sensitivity for stable data transmission
- Supports configurable RF parameters (frequency, spreading factor, bandwidth)

A compact and cost-effective gateway designed for reliable long-range point-to-point communication using LoRa technology.

Operating on a single frequency channel, it enables direct device connectivity without the complexity of a full LoRaWAN network—ideal for small-scale and controlled IoT deployments.

Processor

Cortex-M4 STM32WL 32bit, RISC 48Mhz

Storage

256KB Flash and 64KB RAM

LoRa

Frequency	IN865 MHz, EU868 MHz
Conductive Power	+27 dBm
Receiver Sensitivity	Down to -137 dBm @SF12, BW 125 kHz
Uplinks channel	1
Antenna	External LoRa Antenna

Cellular

Cellular Bands	LTE FDD B1/B3/B5/B8#B34, B38, B39, B40, B41# GSM-GSM/GPRS/EDGE 900/1800 MHz
SIM Card	Nano SIM Tray
Antenna	External 4G LTE

Power Supply

Power Supply	5V-3A DC @ 3A (5V3A)
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Mechanical Specification

Size	135 x 90 x 65 mm
Mounting Type	Wall Mount
Enclosure	Polycarbonate

Pre-Compliance Certification

WPC Certified LoRa Module

Note: *Estimated based on ideal conditions with limited sources of interference

For more information contact at <https://www.nebulae.io> or sales@nebulae.io

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